

HAOJIE HUANG

Email: huang.haoj@northeastern.edu

Personal Website: haojhuang.github.io

EDUCATION

Doctor of Philosophy | *Computer Science*

Northeastern University

Sep. 2021 – Now

Boston, MA

Master of Science | *Robotics*

Northeastern University

Sep. 2018 – May 2021

Boston, MA

Bachelor of Engineering | *Mechanical Engineering*

Shandong University

Sep. 2013 – May 2017

Jinan, China

RESEARCH AREA

Robot Learning, Geometric Deep Learning, Manipulation, Symmetry / Equivariance, Action Representation, 2/3D Perception, Reinforcement Learning

WORK EXPERIENCE

Applied Scientist Intern

Amazon Robotics

Sep 2024 – Jan 2025

North Reading, MA

- Worked on closed-loop Imitation learning algorithm

Research Scientist Intern

Boston Dynamics AI Institute

Sep 2023 – Jan 2024

Cambridge, MA

- Worked on the machine intelligence and robot learning algorithm

R&D Intern

SharkNinja Corp.

June 2019 – Dec 2019

Needham, MA

- Worked on the hardware(sensors, motors and etc.) and software(control algorithms, path planning and etc.) for robotic vacuums

TEACHING EXPERIENCE

Robotic Science and System (CS5335-Teaching Assistant)

Northeastern University

Spring 2021

Boston, MA

AWARD

Best Paper Award - Honorable Mention on IEEE Robotics and Automation Letters

Vienna, Austria

June 2026

Outstanding Paper Finalist at Conference on Robot Learning (CoRL) 2024

Munich, Germany

Nov 2024

Best Paper Award Finalist on Scaling Robot Learning Workshop of ICRA 2022

Philadelphia, United States

May 2022

PROFESSIONAL CONTRIBUTION

- 2026** **Reviewer:** ICLR, ICML, NeurIPS, ICRA, RSS, CoRL, RAL, TRO. **Invited Talks:** Tsinghua University.
- 2025** **Reviewer:** ICLR, ICML, NeurIPS, ICRA, IROS, RSS, CoRL, RAL, TRO. **Invited Talks:** Worcester Polytechnic Institute (WPI); Amazon Robotics. **Paper Presentations:** ICRA 2025.
- 2024** **Reviewer:** ICRA, IROS, RAL. **Invited Talks:** Next-Gen Robot Learning Symposium, TU Darmstadt. **Paper Presentations:** CoRL 2024.
- 2023** **Reviewer:** ICRA, IROS, RAL. **Paper Presentations:** ICRA 2023. **Program Committee:** Workshop on Symmetries in Robot Learning, RSS 2023.
- 2022** **Reviewer:** ICRA, IROS, RAL. **Paper Presentations:** ICRA 2022 (live demo and talk); RSS 2022; RLDM 2022.

FIRST-AUTHOR PUBLICATIONS

- Action Map Policy: Learning 3D Closed-loop Manipulation via Pixel Classification** Jul 2026
Haojie Huang et al. (Contribution: first author)
• Arxiv Preprint -Under Submission
- Pix2Act Image-Space Manipulation Policies with Equivariant Augmentation** Jul 2026
Haojie Huang et al. (Contribution: first author)
• Arxiv Preprint -Under Submission
- Match Policy: A Simple Pipeline from Point Cloud Registration to Manipulation Policies** May 2025
Haojie Huang et al. (Contribution: first author)
• International Conference on Robotics and Automation, Atlantic, ICRA 2025
- Open-vocabulary Pick and Place via Patch-level Semantic Maps** June 2024
Mingxi Jia et al. (Contribution: co-first author)
• IEEE Robotics and Automation Letters 2025
- Imagination Policy: Using Generative Point Cloud Models for Manipulation Policies** June 2024
Haojie Huang et al. (Contribution: first author)
• Conference on Robot Learning, Munich, CoRL 2024
- Fourier Transporter: Bi-Equivariant Robotic Manipulation in 3D** Sep 2023
Haojie Huang et al. (Contribution: first author)
• International Conference on Learning Representations, Vienna, ICLR 2024
- Language-conditioned Equivariant Grasp** Sep 2023
Haojie Huang et al. (Contribution: first author)
• Arxiv Preprint
- Leveraging Pick and Place Symmetries** March 2023
Haojie Huang et al. (Contribution: first author)
• International Journal of Robotics Research, Volume 43, Issue 4, 2024

Edge Grasp Network: A Graph-Based SE(3)-invariant Approach to Grasp Detection Haojie Huang et al. <i>(Contribution: first author)</i>	Dec 2022
• International Conference on Robotics and Automation, London, ICRA 2023 • Sim-to-Real Robot Learning, CoRL 22 workshop, Auckland • Symmetries in Robot Learning, RSS 2023 workshop, Daegu	
Equivariant Transporter Network Haojie Huang et al. <i>(Contribution: first author)</i>	June 2022
• Robotics Science and System, New York, RSS 2022 • Scaling Robot Learning, ICRA 22 workshop, Philadelphia, Best Paper Award Finalists • Reinforcement Learning and Decision Making, RLDM 2022, Rhode Island	
Graph Attention Shape Completion Network Haojie Huang et al. <i>(Contribution: first author)</i>	Dec 2021
• International Conference on 3D Vision, London, 3DV 2021	

COLLABORATED PUBLICATIONS

Retriever: Composing Closed-Loop Asynchronous Robot Programs Linfeng Zhao et al. <i>(Contribution: collaborator)</i>	Jul 2026
• Arxiv Preprint - under review	
Scale Up Strategically: Learning Compositional Generalization via Bias-Aware Evaluation and Data Collection for Robotic Manipulation Yu Qi et al. <i>(Contribution: collaborator)</i>	Jul 2026
• Arxiv Preprint - Under Review	
Evaluating Reasoning Coherence in Video Generative Models with Text and Visual Hints Yu Qi et al. <i>(Contribution: collaborator)</i>	March 2026
• European Conference on Computer Vision - ECCV 2026	
Equivariant diffusion policy for sample-efficient robotic manipulation Dian Wang et al. <i>(Contribution: collaborator)</i>	Feb 2026
• International Journal of Robotics Research, IJRR 2026	
Generalizable Hierarchical Skill Learning via Object-Centric Representation Haibo Zhao et al. <i>(Contribution: collaborator)</i>	Oct 2025
• IEEE Robotics and Automation Letters 2026	
Accelerating Model-Based Reinforcement Learning Using Equivariance Ahmed Agha et al. <i>(Contribution: collaborator)</i>	Sep 2025
• Northeast Robotics Colloquium - NERC 2025 • preprint - under review - RLC 2026	
Clebsch-Gordan Transformer: Fast and Global Equivariant Attention Owen Howell et al. <i>(Contribution: collaborator)</i>	Sep 2025
• Arxiv Preprint	
BEAR: Benchmarking and Enhancing Multimodal Language Models for Embodied AI Yu Qi et al. <i>(Contribution: senior collaborator)</i>	Sep 2025
• Arxiv Preprint - Under Review - ICML 2026	

Equivariant Goal Conditioned Contrastive Reinforcement Learning <i>Arsh Tangri et al. (Contribution: senior collaborator)</i> • Arxiv Preprint	July 2025
3D Equivariant Visuomotor Policy Learning via Spherical Projection <i>Boce Hu et al. (Contribution: collaborator)</i> • Conference on Neural Information Processing Systems, San Diego , NeurIPS 2025	June 2025
Push-Grasp Policy Learning Using Equivariant Models and Grasp Score Optimization <i>Boce Hu et al. (Contribution: collaborator)</i> • IEEE Robotics and Automation Letters 2025 - RAL 25	April 2025
Coarse-to-Fine 3D Keyframe Transporter <i>Xupeng Zhu et al. (Contribution: collaborator)</i> • Arxiv preprint	Feb 2025
OrbitGrasp: SE(3)-Equivariant Grasp Learning <i>Boce Hu et al. (Contribution: co-second author)</i> • Conference on Robot Learning, Munich, CoRL 2024	June 2024
Equivariant Diffusion Policy <i>Dian Wang et al. (Contribution: collaborator)</i> • Conference on Robot Learning, Munich, CoRL 2024, Outstanding Paper Finalist	June 2024
ThinkGrasp: A Vision-Language System for Strategic Part Grasping in Clutter <i>Yaoyao Qian et al. (Contribution: collaborator)</i> • Conference on Robot Learning, Munich, CoRL 2024	June 2024